Skills & Connectors

Lecture 8 Assignment Report

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AI Tool Used: Claude.ai (skill-creator skill, Gmail Connector)

Course: The AI Agent Factory Skills & Connectors Crash Course

Task 1 Build Your First Real Skill: Flashcard Maker

Project Title & What It Does

Flashcard Maker a reusable Skill that generates study flashcards in a clean Question-Answer format for any topic the user names, without requiring the user to paste in notes or source material.

AI Tool(s) / App(s) Used

Claude.ai, using the built-in skill-creator skill to build and package the Flashcard Maker skill.

Prompts Used (and Refinements)

Initial prompt: "flashcard is good" (selecting the topic/idea for the skill)

Clarifying answers given during the skill-creator interview:

- Input type: Topic name only (the AI generates the content itself, no notes needed)
- Output format: Simple Q&A (Question - Answer)
- Trigger phrase: "make flashcards"

Test prompt (used in a new chat, without naming the skill): "make flashcards on [topic]"

How I Tested / Verified It

I opened a brand-new conversation and typed a plain-language request ("make flashcards on topic:DBMS") without mentioning the skill by name. The skill triggered automatically and produced flashcards in the expected format. I repeated this with a different topic to confirm the format stayed consistent.

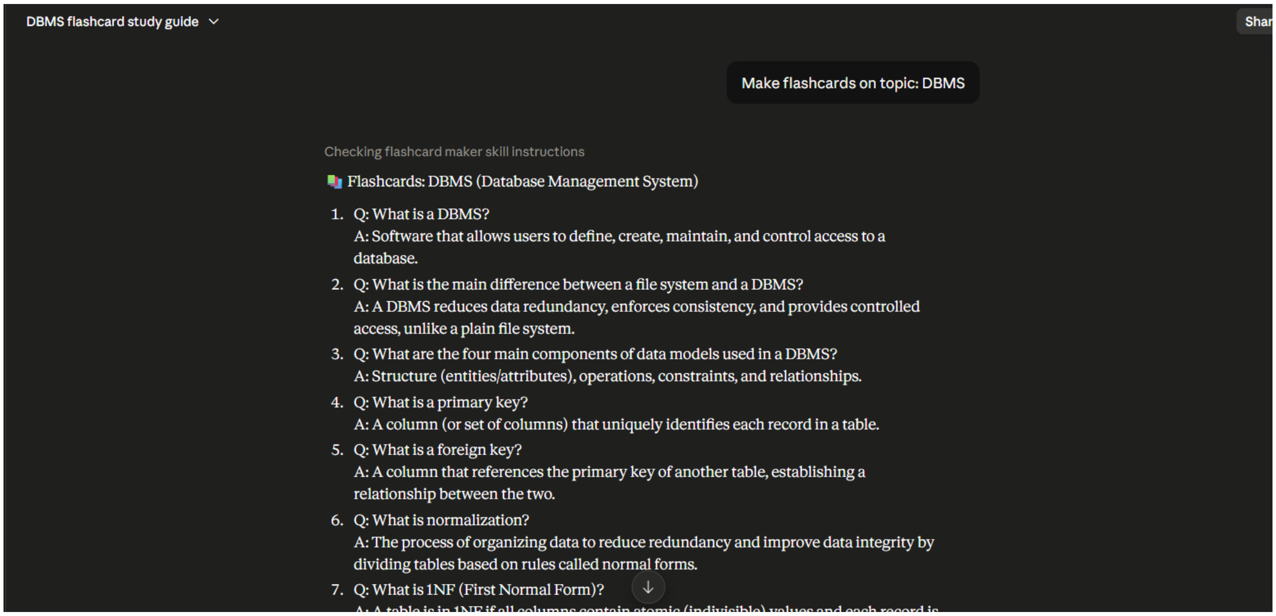
What Worked / What Didn't / Problems Faced

- Worked: The skill triggers reliably from natural language without needing the skill name mentioned.
- Worked: Output format (heading + numbered Q&A pairs) stayed consistent across different topics.
- Note: For very broad topics, the skill focuses on the most commonly important facts rather than trying to cover everything this is expected behavior, not a bug.

Why I Chose This Skill / How It Helps My Daily Life

I chose this because flashcard-based revision is something I do regularly while studying, and manually formatting Q&A sets every time was time-consuming. This skill lets me simply name a topic and instantly get a structured, review-ready flashcard set something I plan to keep using well beyond this assignment.

Screenshot:



Task 2 Connect One App, Read-Only: Gmail

Project Title & What It Does

Connected Gmail to Claude with read-only access and retrieved real email data through the connector no manual copy-pasting involved.

AI Tool(s) / App(s) Used

Claude.ai, connected to Gmail via Claude's Connectors settings.

Prompts Used

Prompt: "summarize my most recent email"

How I Tested / Verified It

After connecting, I asked Claude to summarize my most recent email. Claude retrieved the actual email from my inbox (a Google security alert confirming the connector authorization) and summarized it correctly. I then manually checked my Gmail inbox and confirmed the same email was there, matching Claude's summary exactly.

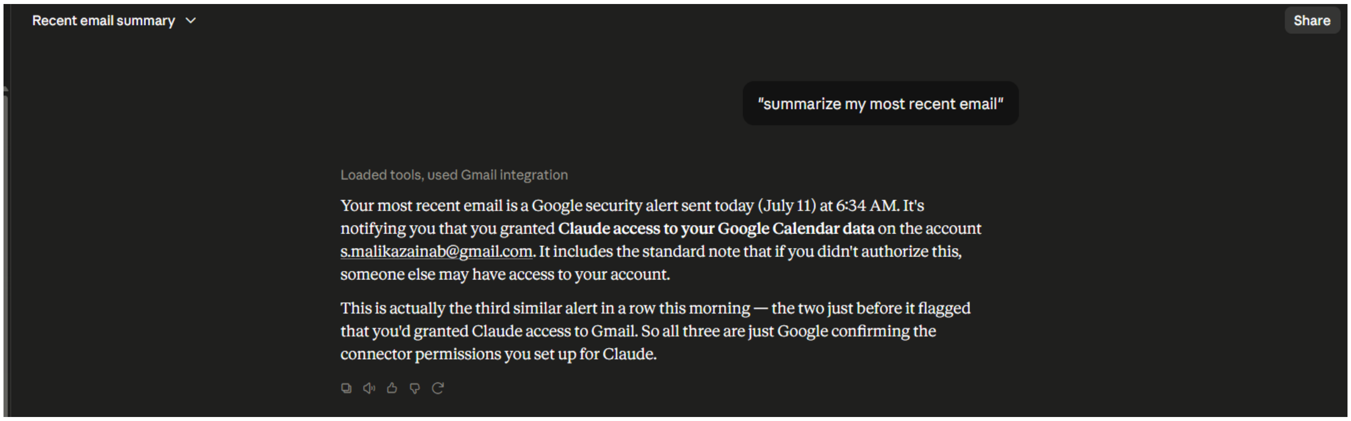
What Worked / What Didn't / Problems Faced

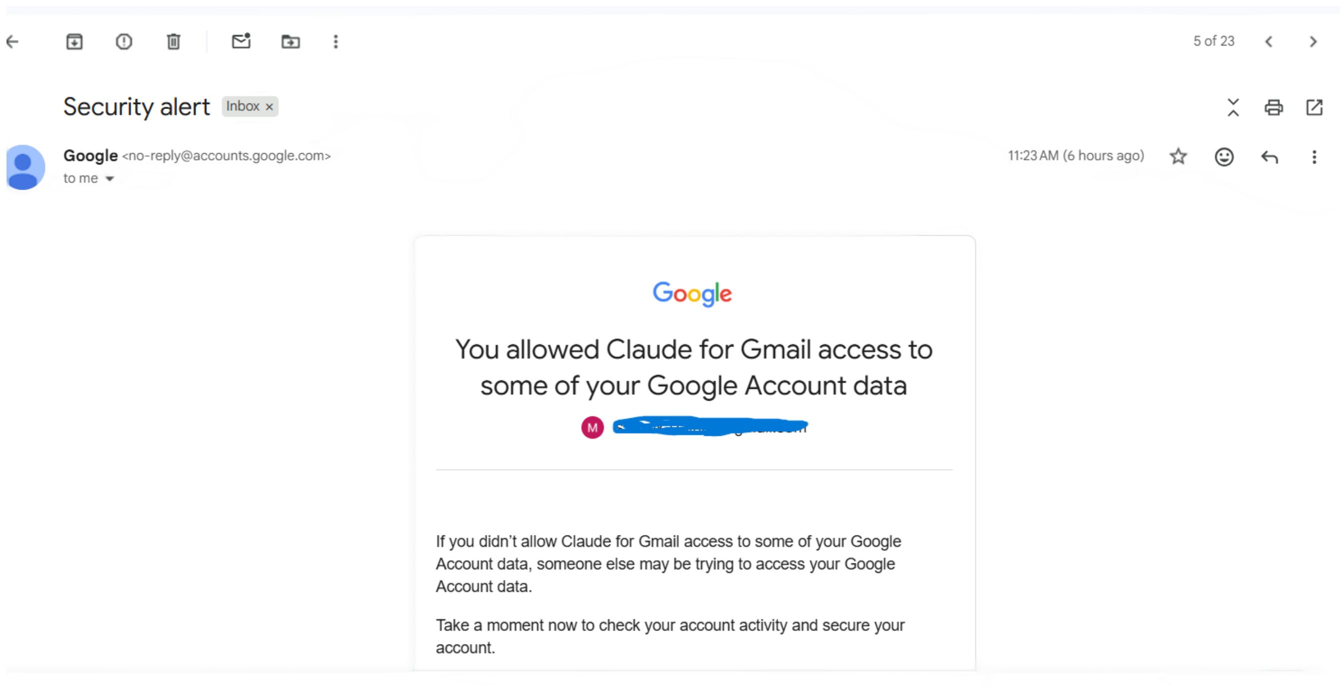
- Worked: Read-only access functioned correctly Claude could search and read emails but had no send/delete capability.
- Problem: In the same chat session where the connector was first authorized, the Gmail tools did not immediately appear in that conversation — this required starting a fresh chat for the tools to become available. This turned out to be expected connector behavior, not a permissions error.

Permission Note

I granted Claude read-only access to Gmail (to view and search my email messages), and I would not need write access (send/draft/delete) because I only want it to retrieve and summarize information, not send or modify anything on my behalf.

Screenshot:





Task 3 Wire Skill + Connector Together

Project Title & What It Does

Combined the Flashcard Maker Skill (Task 1) with the Gmail Connector (Task 2) into a single plain-English request the connector fetched live email content and the skill automatically formatted it into flashcards.

AI Tool(s) / App(s) Used

Claude.ai, using the Flashcard Maker skill together with the Gmail connector.

Prompts Used (and Refinements)

Final prompt: "pull the content from my latest email and make flashcards from it"

Refinement note: The Gmail connector could read the body text of an email but could not read the content of file attachments (PDF/PPTX). To work around this, I sent a test email to myself with the actual lecture content pasted directly into the email body (instead of only as an attachment), so the connector had real text to retrieve.

How I Tested / Verified It

I spot-checked the generated flashcard content against the original email text (and the source lecture material) to confirm the questions and answers accurately reflected the real content, rather than generic or hallucinated information.

What Worked / What Didn't / Problems Faced

- Worked: A single sentence successfully triggered both the Connector and the Skill in sequence.
- Worked: The flashcards accurately reflected the real email content.
- Problem: The Gmail connector cannot read PDF/PPTX attachment contents only plain text typed directly into the email body. This was a useful, real lesson in understanding exactly what a connector's read access does and does not cover.

Screenshot:

Share

pull the content from my latest email and make flashcards from it

Searches for email threads from the authenticated user's Gmail account.

Same email is still the latest — **"Lecture 8"** from 07:11 AM. Here are the flashcards generated fresh from it:

📚 **Flashcards: Requirements Elicitation (Lecture 8)**

1. **Q:** What are the four main iterative phases of the Requirements Engineering Process?
   A: Requirements Elicitation, Requirements Analysis and Negotiation, Requirements Specification, and Requirements Validation.
2. **Q:** What drives Requirements Elicitation?
   A: User needs, domain information, existing system information, regulations, and standards.
3. **Q:** What does Requirements Specification lead to?
   A: The creation of a Requirements Document.
4. **Q:** What does Requirements Validation do?
   A: Validates the document against the process inputs to produce the final Agreed Requirements.
5. **Q:** What are the four key dimensions of Requirements Elicitation?
   A: Application Domain Understanding, Problem Understanding, Business Understanding, and Stakeholder Needs and Constraints.
6. **Q:** What are the four primary stages of the elicitation process?
   A: Objective Setting, Background Knowledge Acquisition, Knowledge Organization,

Task 4 Hand Off

Project Title & What It Does

Proved the Flashcard Maker skill is a genuine, reusable asset (not a one-time trick) by handing the packaged skill file to a friend and having her use it independently, with no explanation from me.

AI Tool(s) / App(s) Used

Claude.ai the packaged flashcard-maker.skill file, shared with and used by a friend on her own account.

Prompts Used

Prompt used by my friend, independently: "make flashcard on topic: Impacts of AI on jobs"

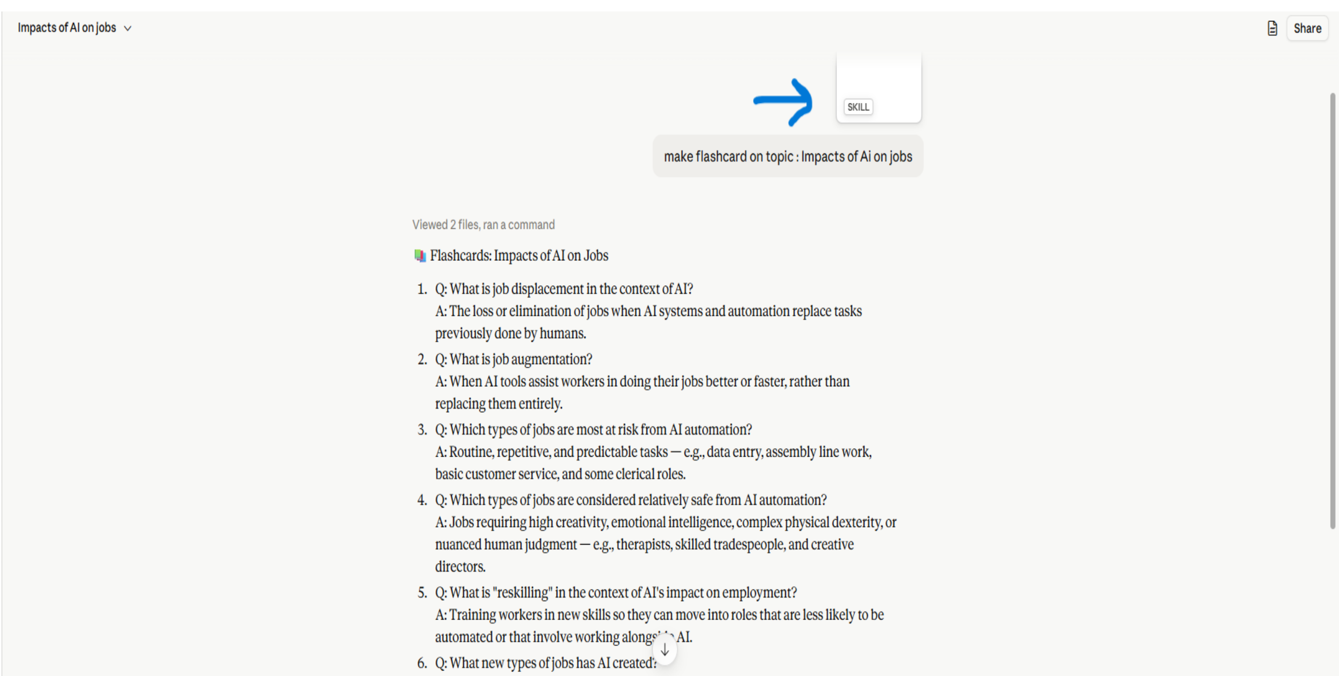
How I Tested / Verified It

I compared my friend's output with the original Task 1 output format: same heading style, same numbered Q&A structure, same tone. She confirmed the skill worked correctly and gave her the flashcards she needed on her own chosen topic.

What Worked / What Didn't / Problems Faced

- Worked: My friend was able to use the skill successfully with zero explanation from me.
- Worked: The skill triggered automatically from her own natural-language prompt.
- Worked: Output format was identical to the original confirming the skill is portable and reusable beyond the original chat.

Screenshot:



Task 5 Audit a Skill Before Trusting It

Project Title & What It Does

Installed and audited Anthropic's official “pdf” Skill (a general-purpose PDF processing toolkit) to practice vetting a Skill for safety before trusting it without needing to read code.

AI Tool(s) / App(s) Used

Claude.ai, auditing the pre-installed official “pdf” skill from Anthropic's public skills directory.

Prompts Used

Prompt: "Explain what the pdf skill does in plain English, and flag anything sensitive does it contact any external server, handle credentials, or send my data anywhere unexpected?"

How I Tested / Verified It

I reviewed the skill's SKILL.md documentation, examined the full folder structure, and scanned all eight bundled Python scripts for network calls, credential handling, and other sensitive operations (searching for terms like “requests”, “urllib”, “http”, “socket”, “api_key”, “token”, and “password”). I then compared the skill's actual behavior against its documented description to check for discrepancies.

What the Skill Does

The pdf skill is a general-purpose toolkit for working with PDF files locally. It can extract text and tables, merge or split PDFs, rotate pages, add watermarks, create new PDFs, fill forms, run OCR on scanned documents, and add/remove password protection all using standard open-source libraries (pypdf, pdfplumber, reportlab, pytesseract) and command-line tools.

Safety Assessment

- External server contact: None found. A full review of all scripts found no network requests, no URLs, and no socket connections. Everything runs locally.
- Credential handling: Minimal and expected a password-protection feature encrypts the local output PDF file only; the password is never transmitted or stored elsewhere.
- Unexpected data transmission: None. Every operation reads a local input file and writes a local output file in the same working directory; no cloud upload, telemetry, or analytics.

Verdict

Safe to enable. The pdf skill performs only local file operations using standard, well-known libraries. It does not access the network, does not exfiltrate data, and does not perform any action beyond what its description promises. There is no reason to distrust this skill.

What Worked / What Didn't / Problems Faced

- Worked: A plain-English explanation and safety flag check was sufficient to reach a confident verdict without reading every line of code.

- No problems faced the audit process (documentation + folder review + targeted code scan) was straightforward and gave clear, conclusive answers.